COMSOL Final Project Report: Topology Optimization of a Bike Frame

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1 Abstract

The primary objective of this project was to minimize the mass of a bicycle frame using computational structural analysis to ensure the frame could withstand the loads applied during use. Utilizing COMSOL Multiphysics, both 2D and 3D topology optimizations were performed to determine the most efficient material distribution. The study evaluated distinct load cases simulating dynamic riding postures, and the ultimate findings detailed the mass reduction percentages achieved without compromising structural integrity.

2 Mathematical Background

2.1 Governing Physics

The structural mechanics of the bike frame under load are governed by standard linear elastic equations. The bike frame begins as a rectangular block with the two supports, one fixed in the bottom left, and a horizontal roller in the top right.

2.2 Topology Optimization Formulation

This study focuses on minimizing the internal strain energy within the bike frame subject to constraints on the total mass available to create the beams of the frame. The solver utilizes the Method of Moving Asymptotes (MMA) to deal with the high number of variables in a complex static loading. The domain of the bike frame essentially became a system of triangles, allowing for an accurate observation of the loading in each beam of the triangle. The optimization algorithm systematically changed the modulus of these beams to find the optimal structural layout.

2.3 The SIMP Method

Material distribution was controlled using the Solid Isotropic Material with Penalization (SIMP) method. This approach described how COMSOL assigned a continuous density variable ranging from 0 to 1, with 0 being a void and 1 being a solid material, to each element in the mesh. This allowed the solver to cut away material that was found to be unnecessary by lowering its density.

COMSOL's automatic setup had a material factor penalization number of 3. However, the default penalization was overly aggressive in cutting out material, which caused the solver to prematurely remove intermediate densities and yielded uninteresting basic results. To resolve this, the penalization factor was lowered to 2 in order to get a more nuanced bike frame that was more interesting to analyze.

2.4 Constraints

The optimization essentially changed the density model all around the bike frame and then tested the loading setup on the new geometry, looking at how it compared to the goal of minimizing the internal strain energy with an upper limit on the mass fraction. As the solver

tested new geometries, it removed mass from zero-force areas to make the frame lighter while keeping the overall strain energy as low as possible.

The mass constraint was set to be 30% of the original geometry, as this provided the best-looking geometry when performing preliminary tests. This first constraint started with the original rectangular domain, found reasonable increases in deflection and strain energy, and was tested repeatedly to determine if the mass was unreasonably heavy (requiring loosened constraints) or unrealistically thin (requiring tightened constraints).

3 Simulation Procedure

3.1 Boundary Conditions

Boundary conditions dictated how the supports worked, how the physical loading was applied, and how the deflection occurred across the geometry. The starting rectangular geometry had two boundary conditions constraining the motion of the bike frame in all directions. In the bottom left of the bike frame, a fixed pin support acted as a rear wheel that was clamped in place to prevent the bike frame from running away horizontally in the simulation. The other boundary condition was a horizontal roller at the top right corner of the bike frame acting as the head tube of a bike frame that would connect to the front wheel.

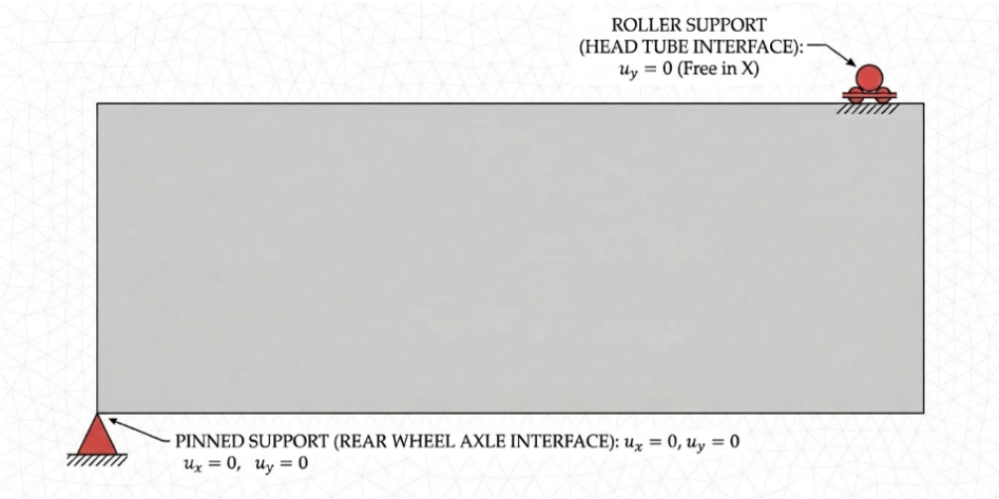


Figure 1: Initial Domain and Boundary Conditions

3.2 Loading Conditions

The bike frame was tested under four separate cases. The first and simplest load case was flat ground with most of the load on the seat. The second load case represented the rider accelerating by pushing harder on the pedals. The third and fourth load cases represented the most extreme loadings of a rider aggressively climbing up a steep incline and coming to an emergency stop by slamming on the brakes.

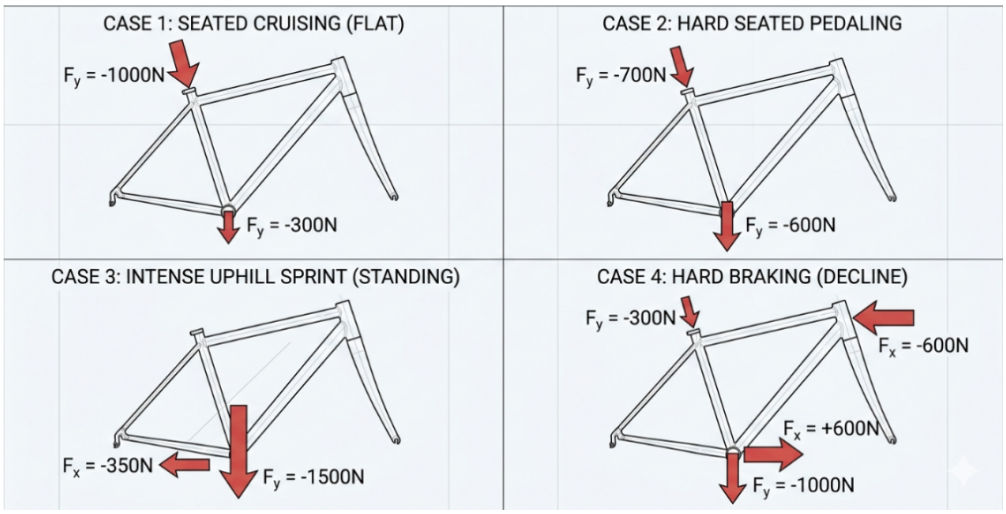


Figure 2: Bike Frame Load Cases

3.3 Optimization Setup

The primary objective of this optimization was to make the lightest bicycle frame possible that could withstand the multitude of loading cases a rider might impose on the frame during typical use. However, running a minimization of the frame mass with a constraint on the internal strain or displacements of different parts of the frame created either an unstable optimization due to the computer's inability to predict the density model's impact on internal strain, or a highly specific solution to prevent displacement in the chosen points on the frame.

The optimization was then modified to minimize the internal strain energy with an upper limit on the mass fraction from the original rectangle. This allowed the computer to predict the constraint more easily and prevented a highly specific local design.

4 Results and Analysis

4.1 Initial Results

The baseline analysis evaluated the initial rectangle results to establish the starting mass, deflection, and internal strain energy. The rectangle originally began as a $0.88 \times 0.43 \times 0.05$ m block of Aluminum 6063, which resulted in a starting mass of 50.7 kg. The rest of the mass of the designs was limited to 30% of this mass, or 15.2 kg.

All four load cases were tested on the initial block with the given mass constraint, essentially giving the block a 30% density throughout. The maximum displacement was 0.026 mm, which was a very minor amount that would be acceptable for the bike frame, but this number was expected to improve with optimization. The rest of the designs were compared to this number as a baseline for displacement.

For the stress distribution, the main factors to consider were how evenly the stress was distributed within the frame and if any parts of the frame reached a dangerous amount

of stress. The stress values were compared to the compressive yield strength and critical buckling stress for Aluminum 6063, which are approximately ∼214 MPa and ∼14 MPa, respectively.

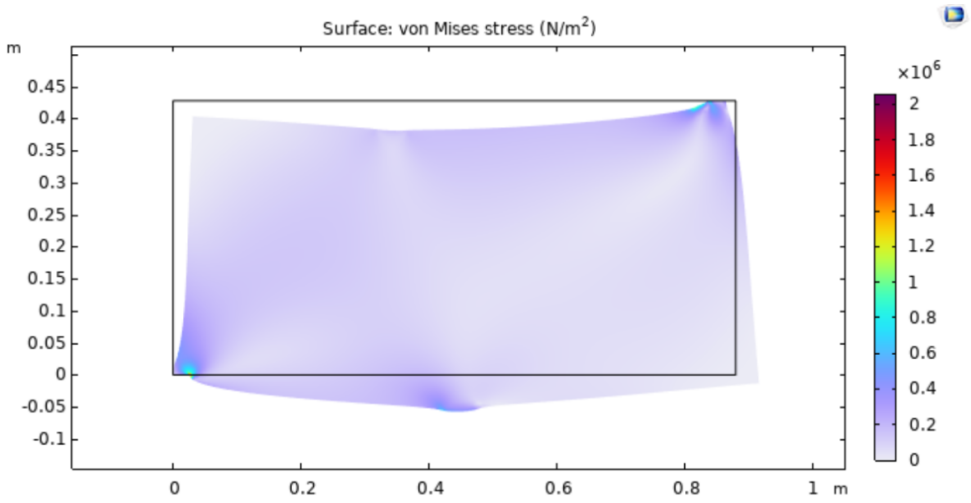


Figure 3: Original Stress Distribution

In Figure 3, it was observed that there was no part of the block experiencing a dangerous amount of stress. There was a slight spike in two minuscule regions at the two supports, but this was such a small area experiencing this stress that it was negligible. The real max stress looked to be around 0.2 MPa in the initial design, very safely under the stress limits.

4.2 Flat Cruising Load Case

In this scenario, the load distribution simulated a rider sitting in their seat while cruising on their bike casually in flat terrain. The load distribution was 1000 N of force directed straight down on the seat and 300 N of force directed down on the crank to simulate the rider casually pedaling.

4.2.1 Flat Cruising Load Case Results

As seen in the results in Figure 4, the optimizer created a very strong rear triangle to direct the load straight into the rear fixed support. This result seemed reasonable as it was very similar to many bike frames on the market today with just a front and rear triangle.

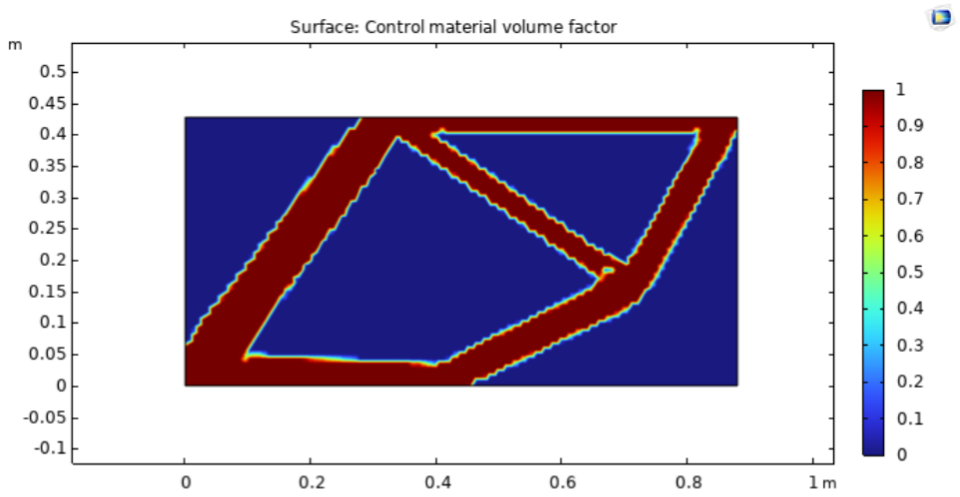


Figure 4: Flat Cruising Topology Results

4.2.2 Stress and Displacement Analysis

The stress analysis results for this load case were presented in Figure 5. The max nominal stress experienced in the frame was ∼0.55 MPa, ignoring mathematical singularities, which was safely under the yielding constraints. The main areas of concern were the upper left by the seat post and the bottom of the rear triangle, as both of these areas experienced about five times more stress than the rest of the frame and would likely be welded joints with a likelihood for weakened rigidity in a physical prototype.

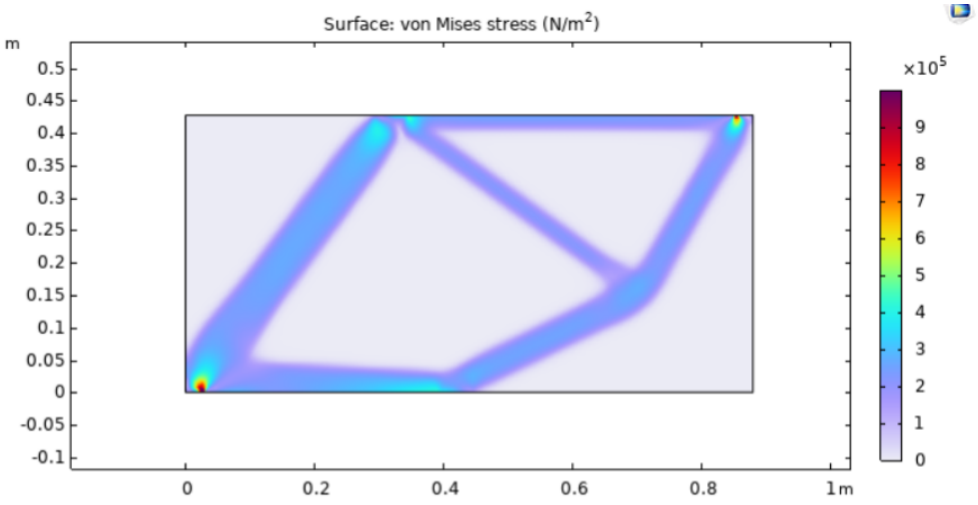


Figure 5: Flat Cruising Stress Distribution

The main aspect of interest, however, was the overall stress distribution, which in this load case was very evenly distributed. It was observed that the theoretical point supports experienced much higher stress spikes than the rest of the bike, which was expected, but

the rest of the structural members were at very low stress, and distributed this load path relatively well throughout the entire frame.

The measured displacement of the bike seat was 0.0053 mm. The topology algorithm successfully reduced the seat deflection to about five times less than the original unoptimized block. This displacement was again very minimal and extremely satisfactory. There was no concern of macroscopic deformation, as it was a negligible amount of strain.

4.3 Fast Pedaling Load Case

In this scenario, a typical use case the bike frame might realistically experience in a slightly more intense loading was simulated. The rider was no longer just sitting with most of their weight in the seat, and the load became a more even distribution between the seat and the crank as the rider began to pedal harder. The load distribution applied was 700 N of force down on the seat and 600 N down on the crank.

4.3.1 Fast Pedaling Load Case Results

As shown in Figure 6, this bike frame was a highly symmetrical design. This made intuitive sense as the redistribution of the loading was moved to be more symmetric between the seat and crank. As the rider was now applying more force to the pedals, the bottom of the frame needed to become more rigid to support this transitioning load.

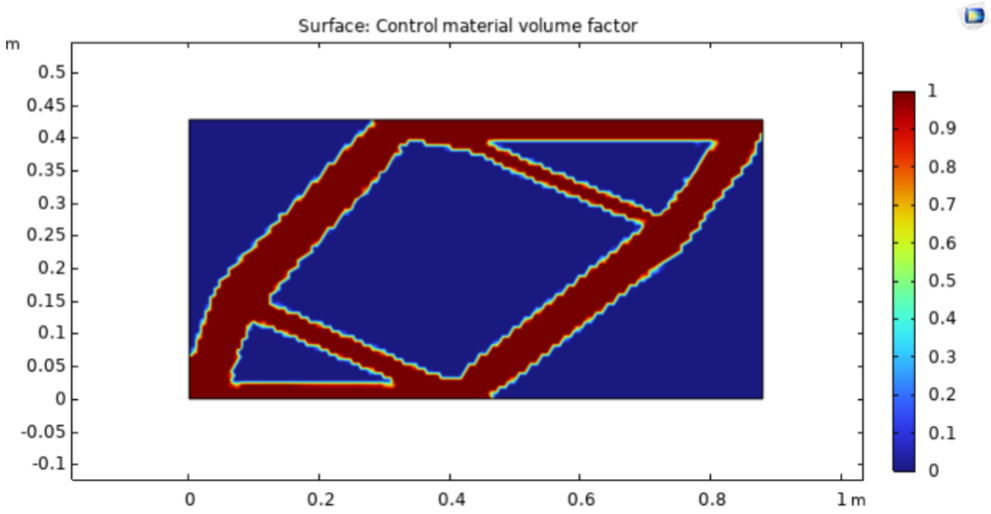


Figure 6: Fast Pedaling Topology Results

4.3.2 Fast Pedaling Analysis

The stress analysis results for this load case were presented in Figure 7. The max nominal stress experienced in the frame was ∼0.35 MPa, ignoring mathematical singularities, which was safely under the yielding constraints. The main areas of concern were the bottom bracket where the crank load was applied and the rear triangle interface, as both of these areas began to experience higher stress concentrations as the rider applied more force to the pedals.

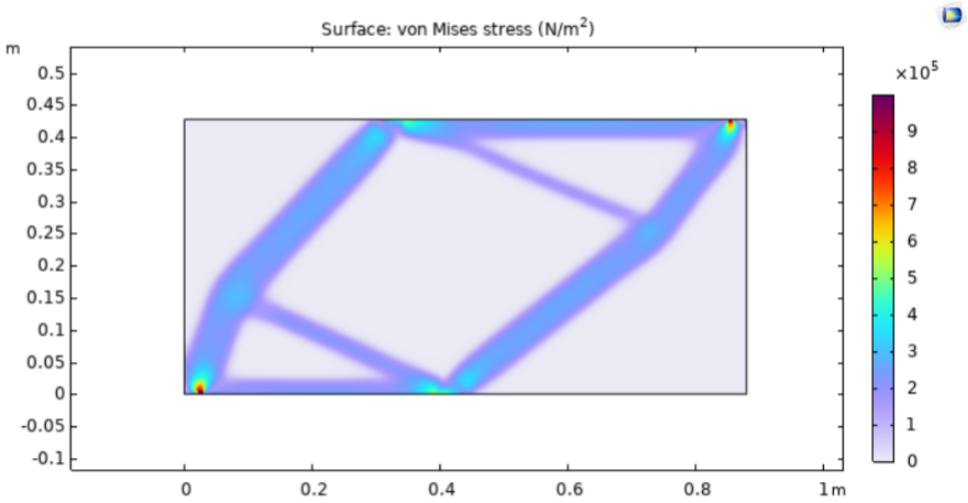


Figure 7: Fast Pedaling Stress Distribution

The overall stress distribution in Figure 7 was still relatively evenly distributed; however, more noticeable spikes were observed at the four corners of the frame. This was something to try to mitigate; however, the overall distribution was still acceptable in this load case.

The displacement of the bike seat was also measured, which was 0.0046 mm. This result was nearly six times less than the original displacement, even beating the first load case. While this displacement was extremely satisfactory, the local displacement was less relevant in this load case compared to the first because the displacement was expected to become smaller as the loading transitioned onto the crank and off of the seat.

4.4 Intense Uphill Sprint Load Case

This load case simulated a rider getting fully out of their seat to ride up an intense incline and pushing very hard on the pedals. This was the only case where the rider's weight force was increased to replicate the strong dynamic forces generated from hard pedaling. The force distribution was 0 N down on the seat, 1500 N down on the crank, and 350 N acting in the negative x direction on the crank. Because the rider was fully off of the seat, it created a very high dynamic load on the bike pedals. Also, due to the average steepness of an intense bicycle climb being about 10-15 degrees, a portion of the force was applied acting towards the rear of the wheel to properly simulate the effect of gravity when climbing at this angle.

4.4.1 Uphill Sprint Load Case Topology Results

This result was slightly unrealistic due to the complexity of the load distribution that occurred from a rider applying lots of torque onto the pedals while accelerating up an incline. The biggest glaring issue was the lack of a bottom on the rear triangle. The main cause was that, in reality, the back tire would need to exert a horizontal force to drive the bicycle up the incline, but with the rear wheel being a fixed support, no loading could be added to the rear triangle. Instead, the algorithm just continued to beef up the downtube of the bike and relied heavily on the front triangle to support all of the forces on the frame.

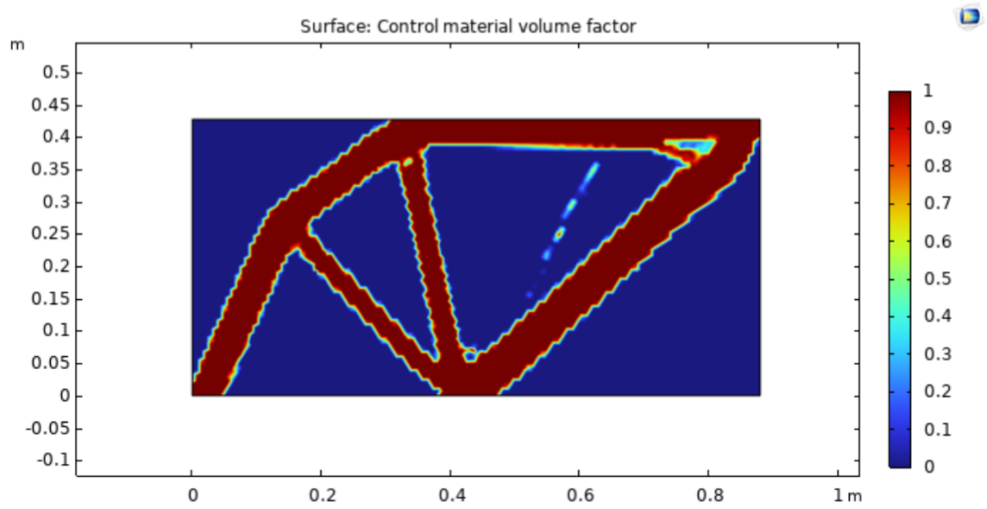


Figure 8: Uphill Sprint Topology Results

4.4.2 Uphill Sprint Load Case Stress Analysis

The stress analysis results for this load case were presented in Figure 9. The max nominal stress experienced in the frame was ∼0.6 MPa at the corners of the front triangle, which was still safely under the yielding constraints. The overall stress was also visibly much higher, which was expected as there was more force acting on the frame.

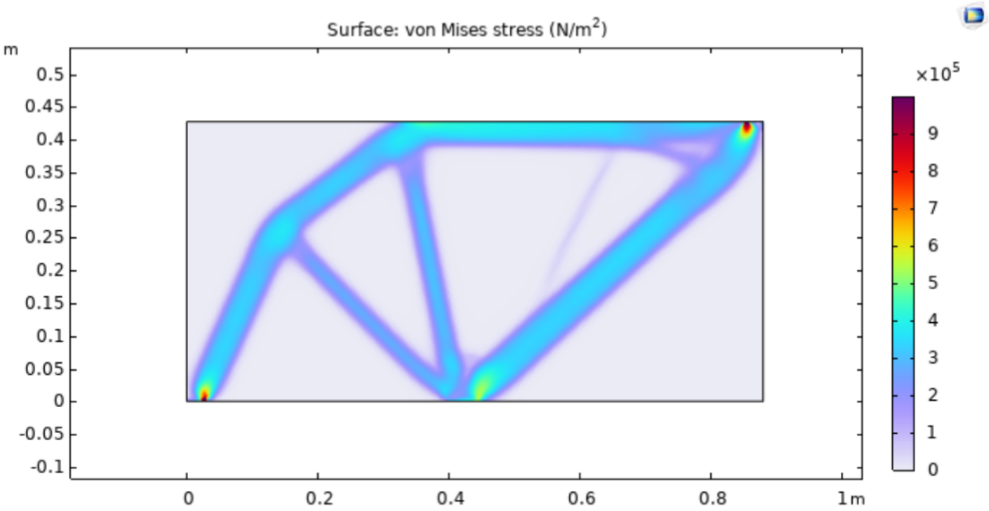


Figure 9: Uphill Sprint Stress Distribution

The main aspect of interest, however, was the overall stress distribution, which heavily favored the front half of the frame. It was observed that the theoretical point supports experienced much higher stress spikes than the rest of the bike, which was expected. However, unlike the previous scenarios, the nominal stress was naturally higher and concentrated heavily within the front triangle. With the rider entirely out of the saddle and applying high

dynamic forces directly to the crank, the front of the frame was forced to act as a primary torsion box, distributing this stress through the down tube.

The measured displacement of the structure was 0.0046 mm. The displacement was again about six times less than the original unoptimized block. This displacement was still quite satisfactory, and it made sense as the loads were being taken off the seat entirely, but the overall loading had increased.

4.5 Hard Braking Load Case

In this final load case, a rider slamming on the brakes as if they were coming to an emergency stop was simulated. The resulting force distribution was 300 N down on the seat, 1000 N down on the crank, 600 N acting in the positive x direction on the crank, and a massive 600 N acting in the negative x direction on the head tube. The rider had to stand firmly on the pedals to brace for the sudden stop. The front wheel would also be pushing aggressively to the left as the front brake was applied, which caused the horizontal forces.

4.5.1 Hard Braking Topology Results

In Figure 10, it was observed that in this load case, the algorithm made the top tube and upper arm of the rear triangle take a large majority of the available material to help hold the head tube in place under the intense braking forces. The bottom bracket became much thinner as it was less of a priority than the top of the frame, but the bottom bracket now had three connections to the top and left of the frame to help support the horizontal forces from the rider pushing back on the pedals.

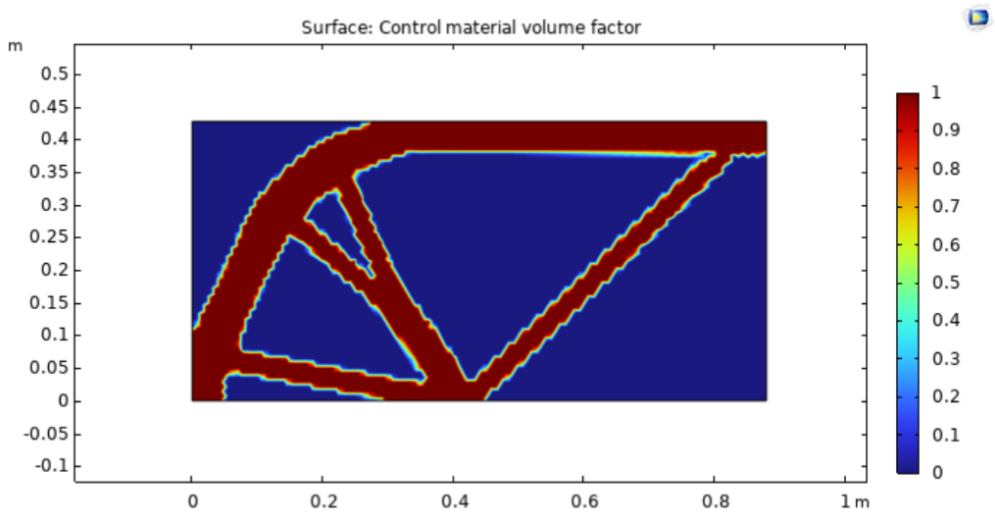


Figure 10: Hard Braking Topology Results

4.5.2 Hard Braking Stress and Displacement

The stress distribution in this load case was higher than other load cases again because of the increase in the amount of forces on the frame. The max stress in this load case was

~0.55 MPa, mostly throughout the top tube, but the biggest stress spike was in the rear triangle.

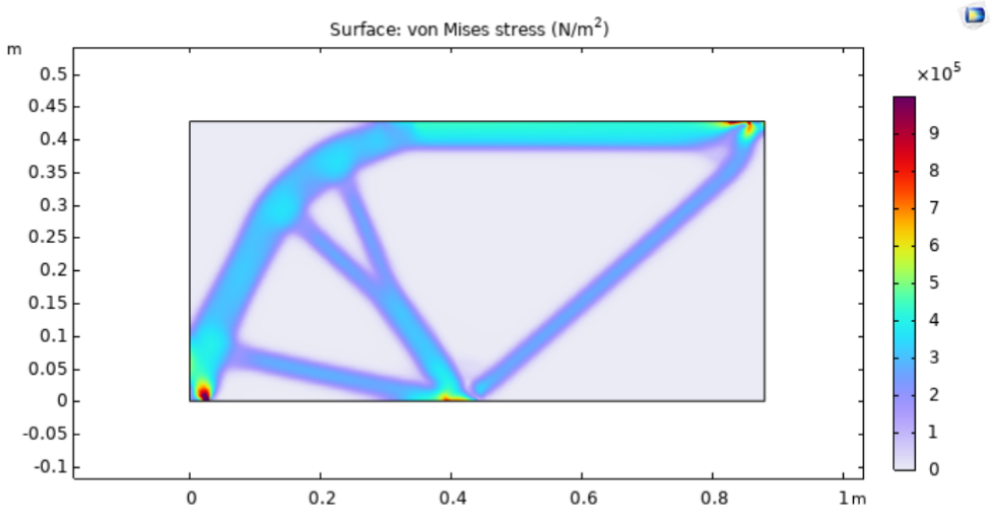


Figure 11: Hard Braking Stress Distribution

The most concerning part of this stress distribution was the high spike in the rear triangle, likely due to the horizontal forces from braking. However, this stress spike was still relatively small and was not overly concerning. The most interesting part of this distribution was how the curved top part of the frame visibly flowed the stress from the horizontal force on the head tube down into the fixed support, indicating that the frame was successfully transferring the braking force away from the front fork and into the rear supports.

The measured displacement of the seat was 0.0050 mm. The algorithm again created a frame with displacement about five times less than the original block. This was very satisfactory, and plastic deformation was not a concern in this load case.

4.6 Summary

In order to truly optimize the bicycle frame, it was necessary to look at the results from all the different load cases rather than a single static scenario. Biking is a highly dynamic activity, and a frame was needed that could undergo all of the different loading phases. For example, as the results showed, a frame optimized exclusively for a seated rider cruising on a flat road would likely fail the moment the rider stood up to sprint or heavily applied the brakes, because the load paths completely changed.

The most important factors from all the load cases were having a strong top tube for braking, as this was a common and intense loading case, as well as enough support on the bottom bracket in all directions—but primarily upwards—to support the intense loading during braking and hard pedaling when the rider was fully off the seat and dynamic forces were involved. The seat surprisingly did not require that much support even though it was such a core piece of a bike frame, as it only experienced mostly static loading at low magnitudes.

5 Three Dimensional Optimization

For this three-dimensional analysis, a setup nearly identical to the 2D optimization was run, utilizing the same boundary conditions, but only a single load on the seat. However, expanding into the 3D domain drastically increased the number of variables, making the solver highly computationally expensive. Even after applying a much coarser mesh and using the simplest loading possible, this optimization took approximately 8 hours to run before it timed out with 100 iterations.

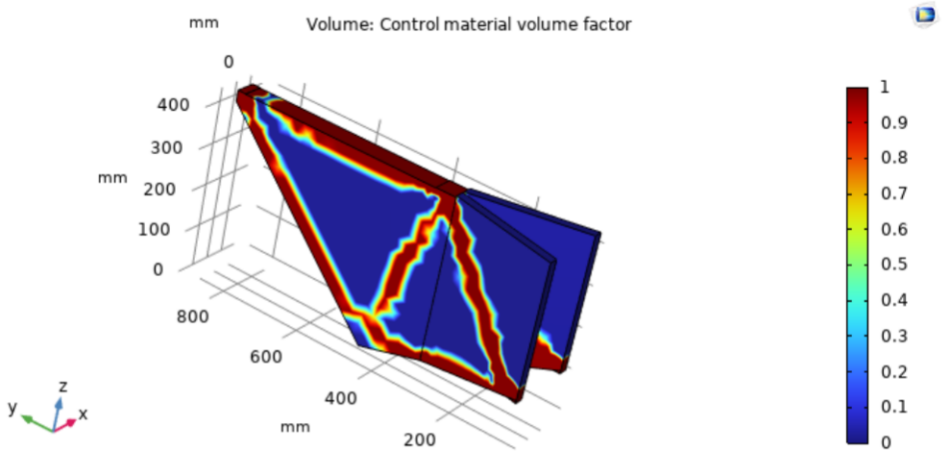


Figure 12: 3D Topology Result

The much coarser mesh resulted in a blocky, uninteresting design of three simple triangles before the optimization timed out. Furthermore, in the 3D optimization, if a more precise optimization had been run over an extremely long time, it might have resulted in a highly complex, organic design that was not realistically machineable. Due to this computational bottleneck, this report focuses on the 2D optimization. If a 3D design was necessary, it would be easiest to extrude the 2D design, or design a pipe frame based on the outline of the 2D optimization results.

6 Shortcomings and Future Analysis

6.1 Future Optimizations

While this study generated highly effective topologies for each individual load case, there were inherent limitations. As noted in the 3D Flat Case, computational constraints required the use of a coarse mesh, which obscured the fine topological details necessary for high-fidelity manufacturing. Future analysis could be done in 3D with a much finer mesh and several days of optimization runtime to find a more accurate 3D structural design. In both 2D and 3D, a sweep could have been run testing all of the different load cases at the same time to ensure the bike was truly versatile, but this again required more computation time and added complexity to the optimizations.

6.2 Pipe Frame Analysis

Additionally, this optimization created a bike frame made of solid beams that experienced very low stress in comparison to the critical buckling and yield stress of aluminum. However, it should be noted that nearly all bike frames are hollow, not solid beams. Therefore, in the future, a bike frame of pipes could be designed based on the geometry from the 2D optimization. This new 3D frame could have a simple finite element analysis performed on it to ensure that the design was safe with a given pipe diameter, or another parameter optimization of this design could be run that modified the diameters of each pipe to determine the optimal pipe sizing for every single member.

7 Conclusion

The primary goal of this study was to create a stiff, lightweight bicycle frame that was resilient across multiple different loading conditions. The optimizations successfully generated a unique, highly efficient topology for all four of the load cases tested, proving vastly more effective than the original design. In all cases, the algorithm strategically reallocated the available material to create a bike frame that experienced safe, comparable stress and significantly less displacement.

By satisfying the 30% mass constraint, the overall bike frame weight was cut from 50.7 kg to 15.2 kg, while still keeping the internal stress and deflection of the structure to very low values. This proved that the MMA solver successfully removed redundant material without compromising the integrity of the frame. While this study was ultimately limited by static approximations of dynamic forces and computational bottlenecks, the results were highly impressive, demonstrating that the solver is capable of creating incredibly effective structural designs.